

NFTs, Blockchain, and Ethereum

Learn the rules and the basics so you can think beyond them

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Based on the April 2022 workshop

Why are you here?

What would you like to understand or make?

By the end, you should be able to:

- Explain how a token relates to its artwork and metadata.
- Describe what a wallet asks you to authorize.
- Separate evidence about a project from its promises.

What is an NFT?

An NFT is a token that a blockchain can distinguish from other tokens.

A project may associate it with an artwork, a membership, or another use.

The token record and the thing it represents are separate parts of the system.

Wallets and accounts

Address

An address identifies an account.

Wallet

A wallet provides an interface for using accounts.

Authorization

Authorization lets the account take an action.

Recovery details and private keys must stay private.

What does the blockchain record?

Ethereum maintains shared records that network participants check against protocol rules.

For an NFT, those records can include its creation, transfers, and current owner account.

An explorer makes those records easier to inspect.

Smart contracts

A smart contract is a program on the blockchain.



Its code can define who may mint, transfer, or change parts of a collection.

The project's website is one interface to those rules.

Minting and transferring

Minting creates a token in the collection's records.

Transferring changes its recorded owner.

A typical mint involves:

1

Reviewing the request in a wallet.

2

Authorizing and submitting a transaction.

3

Checking whether it succeeded on the intended network.

Gas and transaction costs

Gas measures computational work.

Transaction fee

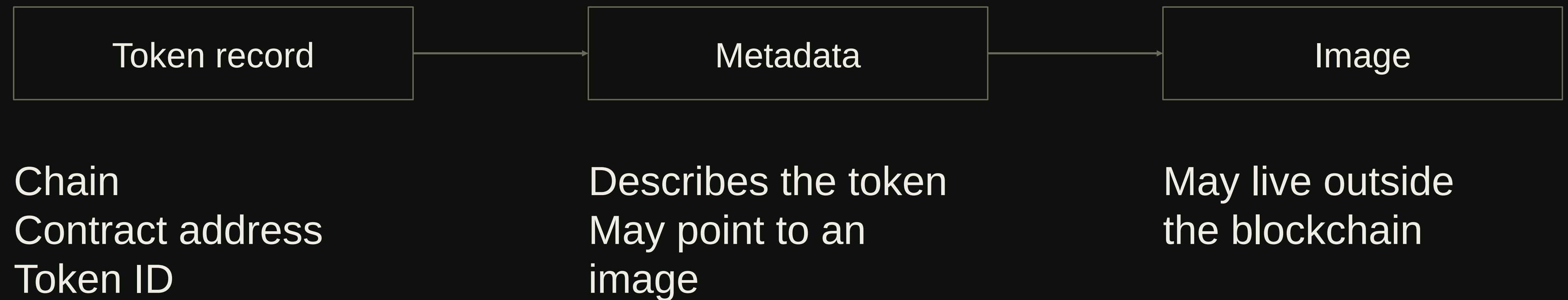
= gas used × effective price per gas

The network fee is separate from an NFT's purchase price.

A failed transaction included onchain can still cost gas.

Artwork, token, and metadata

For a typical ERC-721 NFT:



Where does each part live? Who can change it?

Conceptual example

Provenance and authorship

Provenance concerns an object's origin and history.

A token's history can show its mint and transfers.

The connection to an artist still needs evidence.

What links this contract to the person claiming to have made the work?

What does ownership include?

Token ownership, copyright, and access benefits are separate questions.

Check the terms:

- What may the holder do with the artwork?
- Who provides any promised access or service?
- What happens when the token changes hands?

What am I authorizing?

Connecting, signing, approving, and transferring can do different things.

Before authorizing:

- Check the site and requested action.
- Identify the assets and permissions involved.
- Stop if the request is unclear.

A request with no gas fee can still carry risk.

Evidence, promises, and unknowns

Illustrative project exercise

A collection page shows artwork, a contract link, a price, and a promise of future events.

Which statements can you verify from that page?

What would you inspect next?

How to build a community

What would people want to do together here?

- Give people a meaningful way to participate.
- State what the project can deliver.
- Explain who handles questions and problems.
- Make changes and limitations visible.

Contribute something you would want to exist.

What can you explain now?

- 1 How does a token relate to its image?
- 2 What might a wallet signature authorize?
- 3 What does a project's contract prove, and what remains uncertain?

What would you investigate before participating?

Sources and further reading

[Ethereum.org: NFTs, wallets, transactions, gas, and security](#)

[ERC-721: the token standard](#)

[IPFS Docs: content identifiers and persistence](#)

[U.S. Copyright Office: NFTs and intellectual property](#)

Original workshop: April 2022
Revised material: September 2026

Appendix A: Ethereum since April 2022

April 2022

Original workshop

September 15, 2022

Ethereum moves to
proof-of-stake

Wallet capabilities and transaction formats have continued to develop.

Historical explanations need dates when reused.

Appendix B: Accounts and contract execution

Ethereum supports accounts controlled through keys and accounts governed by contract code.



A transaction can call a contract, which may call other contracts.

Execution uses gas and can fail if required conditions are unmet.

Appendix C: Gas worked example

Hypothetical Ethereum mainnet example

50,000 gas × 12 gwei per gas

= 600,000 gwei

Network fee: 0.0006 ETH

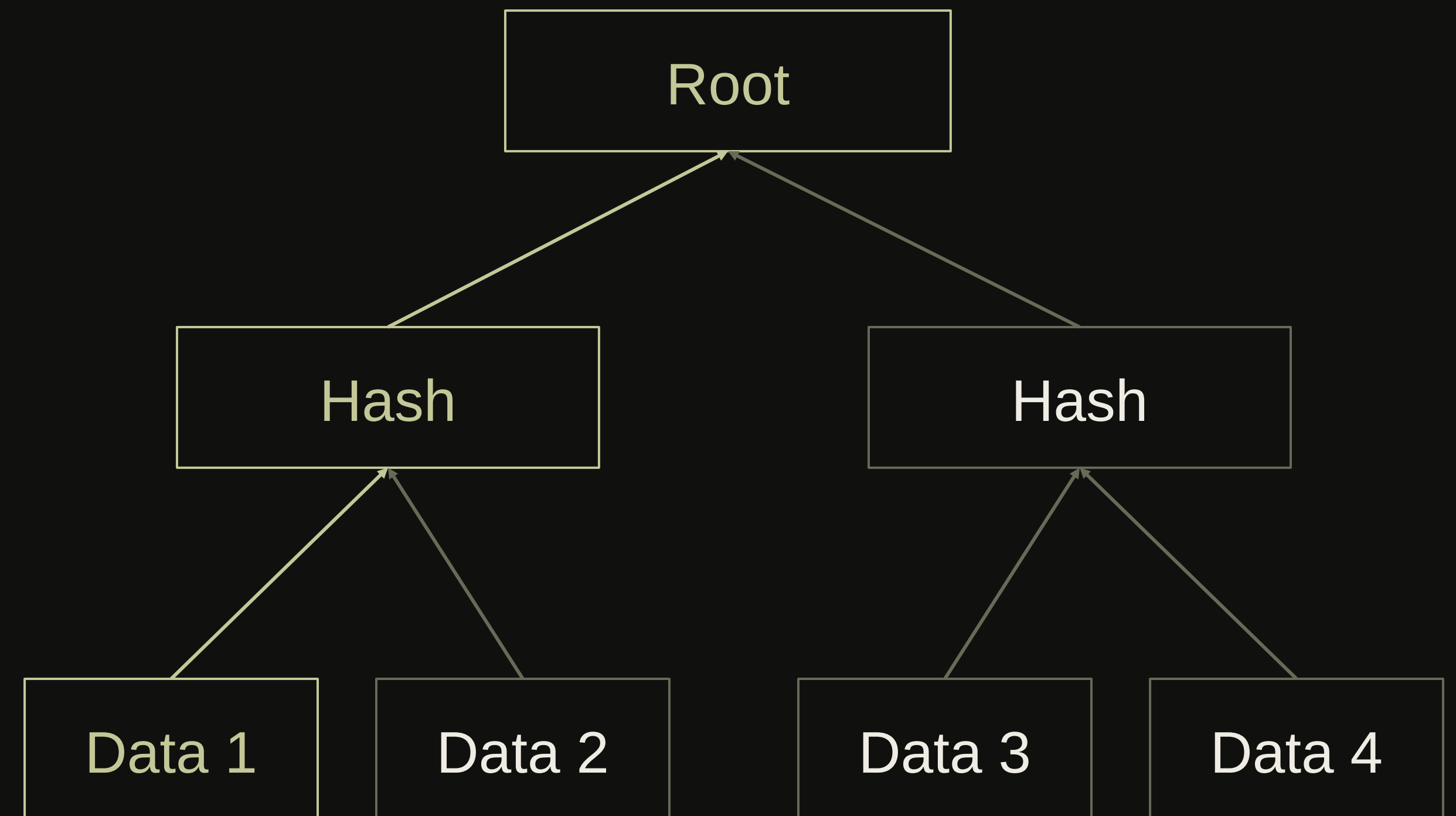
Any purchase price is additional.

Appendix D: Hashes and Merkle proofs

A hash summarizes data in a fixed-size value.

A Merkle structure combines hashes into a root.

A proof can establish that data belongs under a known root without sending the entire dataset.



Conceptual example